

Spanning Tree Protocol

- Topics covered:
 - Network Redundancy
 - Spanning Tree Protocol

Network Redundancy

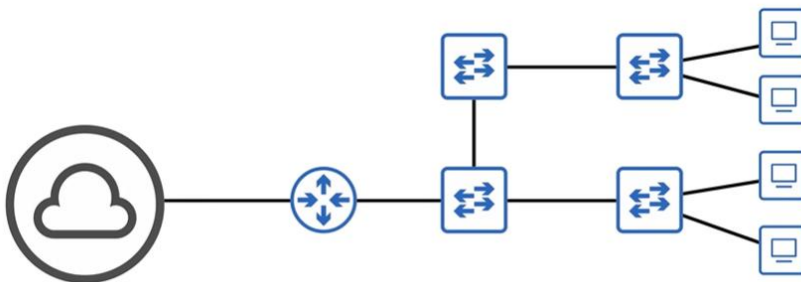
◆ Redundancy in Networks

Redundancy means having backup devices and links in a network.

- Modern networks must run 24×7
- Even small downtime can cause business loss
- If one device or link fails, another path must be available

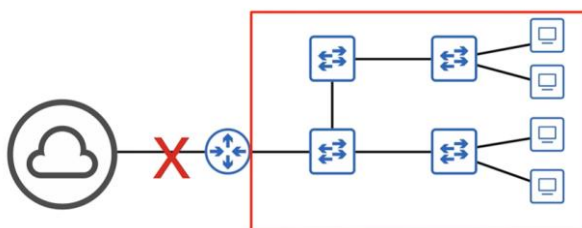
Goal: Ensure the network continues to work even during failures.

◆ Network Without Redundancy (Bad Design)

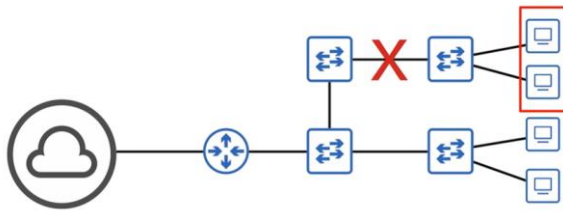


Problems:

- Router failure → Entire network down

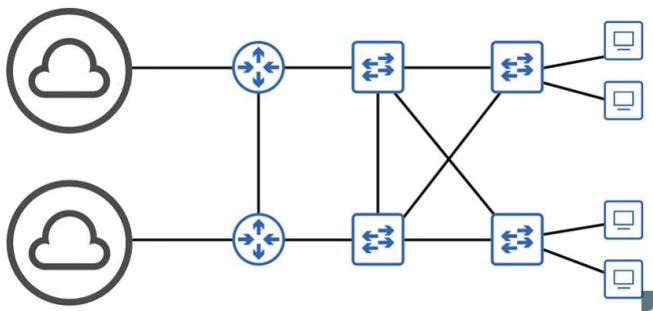


- Switch failure → LAN + Internet down



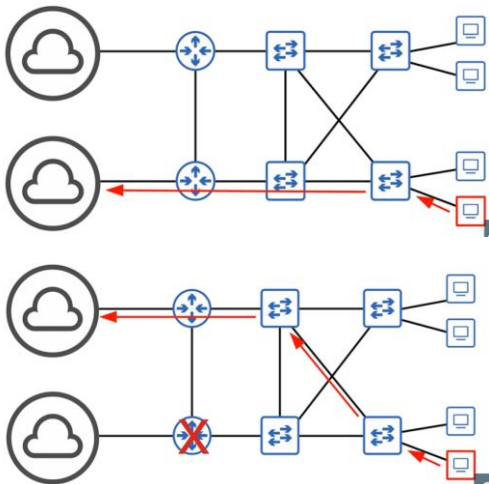
- Many single points of failure

◆ Network With Redundancy (Better Design)

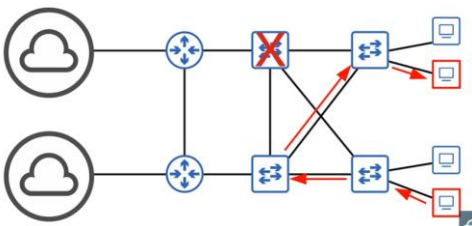


Advantages:

- Backup router available

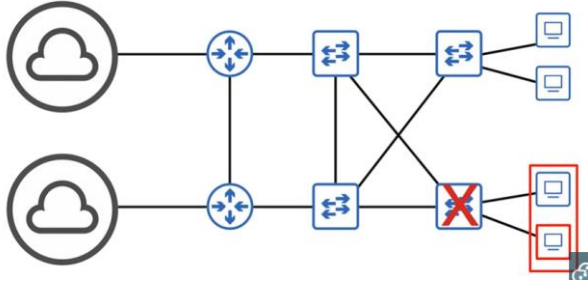


- Backup switch path available



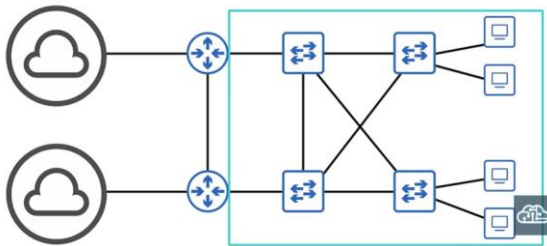
- Network survives failures

◆ Limitation of Redundancy



- PCs usually have **only one NIC**
- If access switch fails → PCs lose connectivity
- Servers often have **multiple NICs** for redundancy

◆ SPANNING TREE AND NETWORK REDUNDANCY



Spanning Tree is a Layer 2 protocol.

It works inside the LAN, not for routing to the Internet at Layer 3.

It enables redundant Layer 2 networks.

◆ WHY REDUNDANCY IS IMPORTANT

Redundancy means having multiple paths between network devices.

If one connection fails, another path can be used.

This makes the network reliable and always available.

◆ THE MAJOR PROBLEM WITH REDUNDANT LAYER 2 NETWORKS

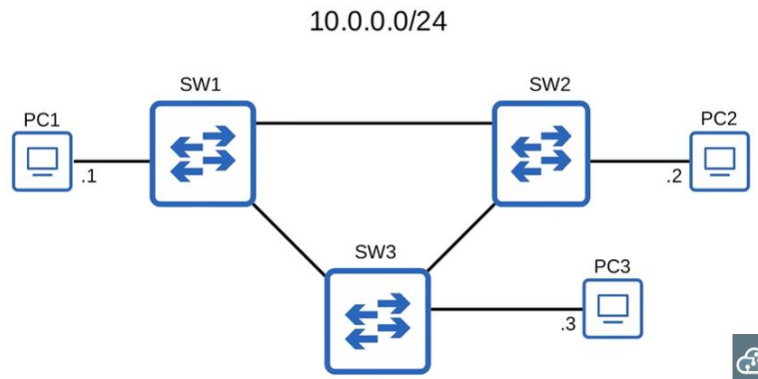
Without Spanning Tree, redundant networks have a major problem.

This problem can destroy the network.

◆ INTRODUCTION TO BROADCAST STORMS

The problem is called a broadcast storm.

To understand this, we use a simplified topology.



ARP Request Scenario :

PC1 wants to send traffic to PC2.

PC1 does not know PC2's MAC address.

So PC1 sends an ARP Request.

- Destination MAC: FF:FF:FF:FF:FF:FF
- ARP Request is a broadcast frame

How A Switch Handles A Broadcast Frame :

When SW1 receives the ARP Request:

- It floods the frame out **all ports**
- Except the port it was received on

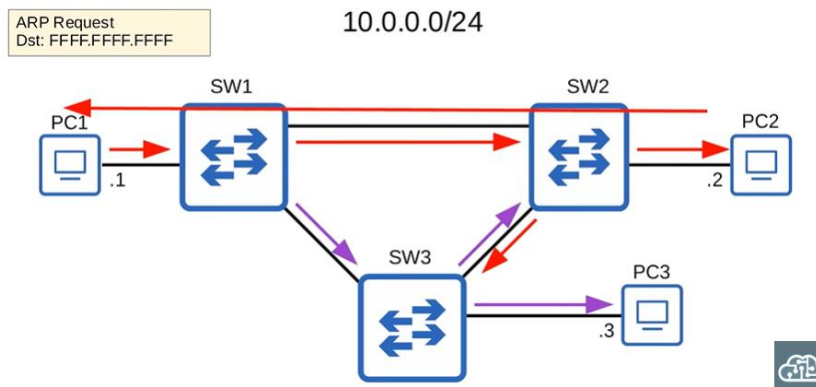
So both SW2 and SW3 receive the frame.

BROADCAST FLOODING CONTINUES

SW2 and SW3 do the same thing:

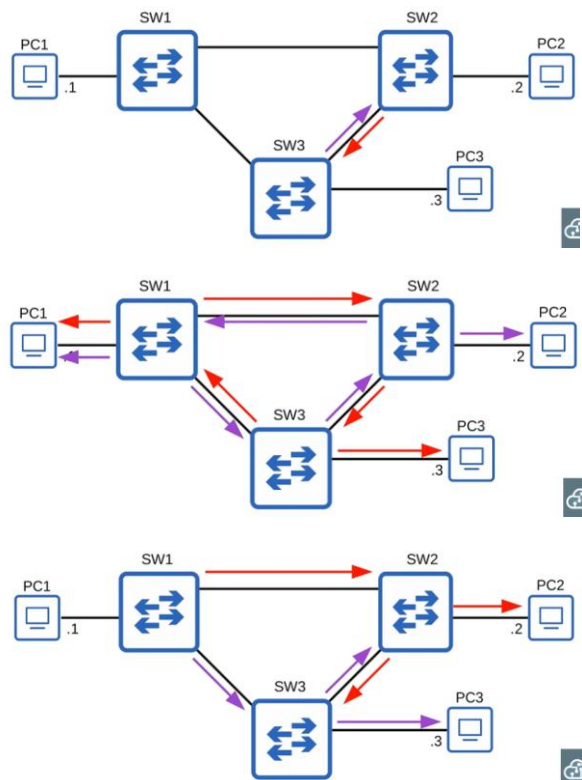
- Flood the frame out all ports
- Except the incoming port

PC2 receives the ARP request and replies with a **unicast ARP reply**.



THE PROBLEM DOES NOT STOP

Even though PC2 replied, the broadcast frames remain in the network.



WHY THE LOOP NEVER ENDS

At Layer 3, IP uses TTL (Time To Live).

TTL prevents infinite loops.

Ethernet at Layer 2 has NO TTL field.

So broadcast frames never expire.

BROADCAST STORM

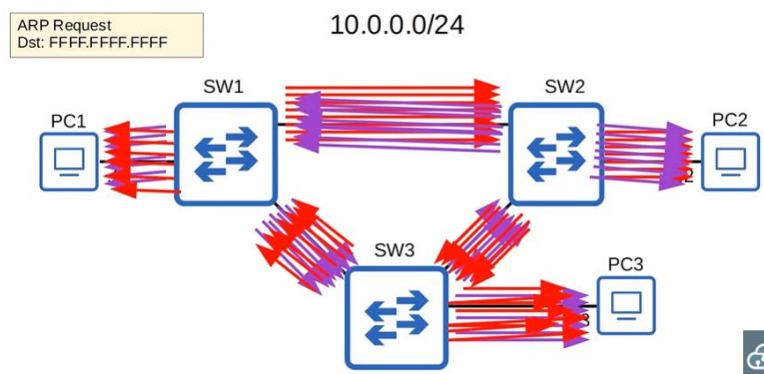
Broadcast frames keep looping and multiplying.

The network becomes:

- Extremely congested
- Unable to carry real traffic

This condition is called a broadcast storm.

NETWORK DURING A BROADCAST STORM



Red arrows = looping broadcasts

Purple arrows = looping broadcasts in the opposite direction

Other Problem MAC ADDRESS FLAPPING

Each time a frame arrives, the switch:

- Learns the source MAC address
- Updates the MAC address table

Because frames arrive on different ports repeatedly:

- MAC table keeps changing
- This is called MAC Address Flapping

INTRODUCTION TO SPANNING TREE PROTOCOL

◆ SPANNING TREE PROTOCOL

Spanning Tree Protocol is the solution.

Classic Spanning Tree Protocol is IEEE 802.1D.

This is the STP version used here.

◆ STP IS VENDOR INDEPENDENT

STP is so important that:

- Switches from all vendors run STP
- Not only Cisco switches

◆ HOW SPANNING TREE WORKS

STP prevents loops by:

- Placing redundant ports into a blocking state

Blocked ports are logically disabled.

FORWARDING STATE:

- Sends and receives normal traffic

BLOCKING STATE:

- Does not forward user traffic
- Only sends and receives STP messages (BPDUs)

◆ WHAT DOES “BRIDGE” MEAN IN STP

Before switches, networks used bridges.

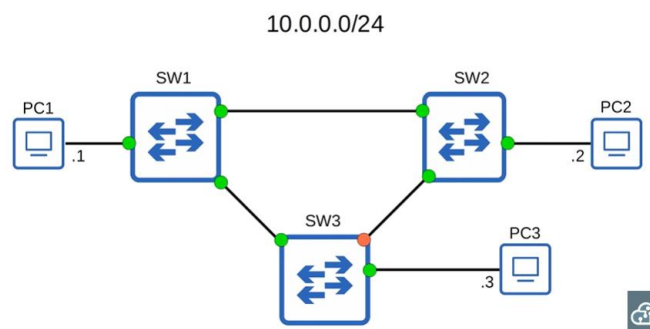
STP still uses the word bridge.

In STP:

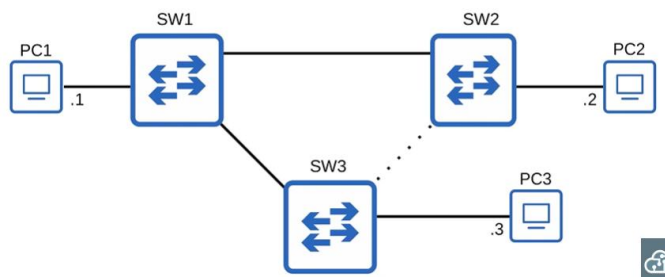
- Bridge = Switch

Bridges are not used today.

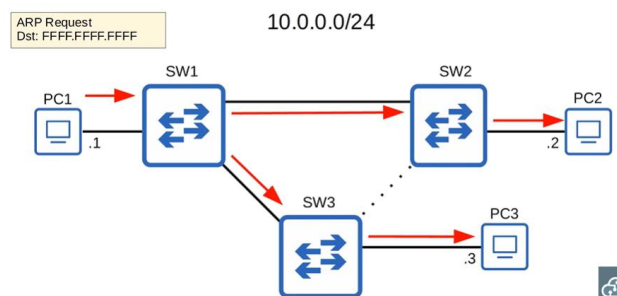
◆ STP TOPOLOGY EXAMPLE



STP APPLY



ARP REQUEST WITH STP ENABLED



PC1 sends the ARP Request.

The frame is flooded only once.

There is no looping.

FAILURE AND RECOVERY WITH STP

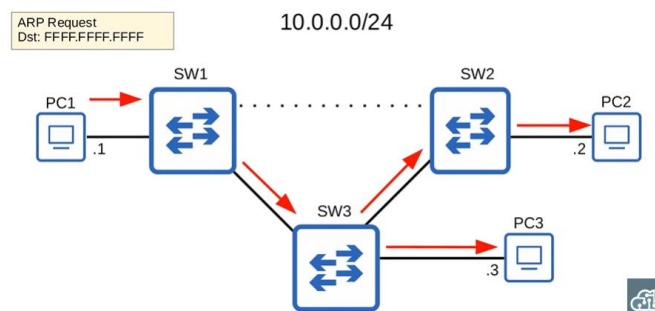
If an active link fails:

BEFORE FAILURE:

SW2 ---X--- SW3

AFTER FAILURE:

SW2 ----- SW3



STP automatically unblocks the backup port.

◆ PURPOSE OF SPANNING TREE PROTOCOL

Spanning Tree Protocol is a Layer 2 protocol.

It is used inside the LAN to allow redundancy while preventing Layer 2 loops.

By choosing which ports should forward and which should block, STP creates one single logical path between all points in the network.

This is how Layer 2 loops are prevented.

Advantage of STP:

- Prevent Layer 2 loops
- Prevent broadcast storms
- Prevent MAC address flapping
- Allow redundant paths
- Automatically recover from failures

◆ HELLO BPDUs AND INITIAL DISCOVERY

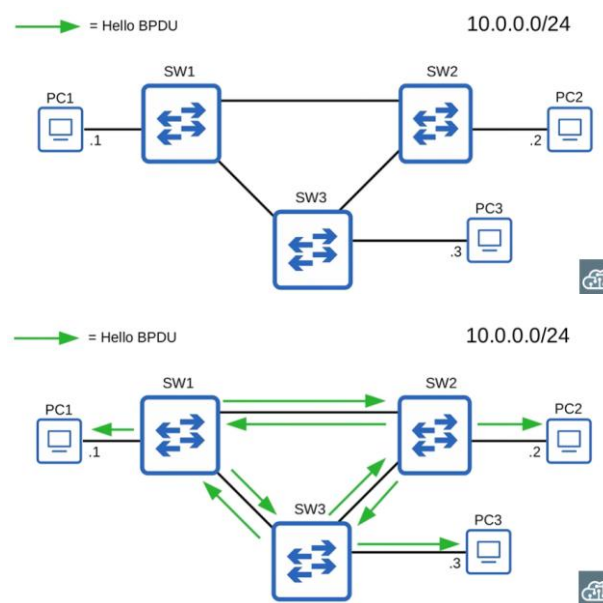
STP-enabled switches send Hello BPDUs out of all interfaces.

The default timer is 2 seconds, so every switch sends a Hello BPDU from every port every 2 seconds.

If a switch receives a Hello BPDU on an interface, it knows that the interface is connected to another switch.

Routers, PCs, and other devices do not use STP, so they do not send BPDUs.

BPDU FLOW (HELLO BPDUs)



Green arrows in the diagram represent Hello BPDUs being exchanged between switches.

Switches use these BPDUs to advertise themselves and learn information about other switches.

◆ ROOT BRIDGE ELECTION

BPDUs are used to elect the root bridge.

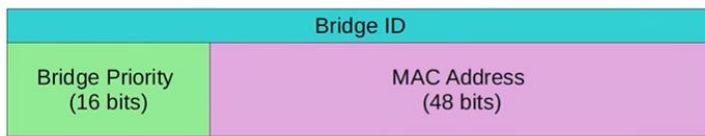
The switch with the lowest Bridge ID becomes the root bridge.

All ports on the root bridge are placed into a forwarding state.

Every other switch must have a path to reach the root bridge.

◆ BRIDGE ID (TRADITIONAL FORMAT)

Originally, the Bridge ID had two parts:

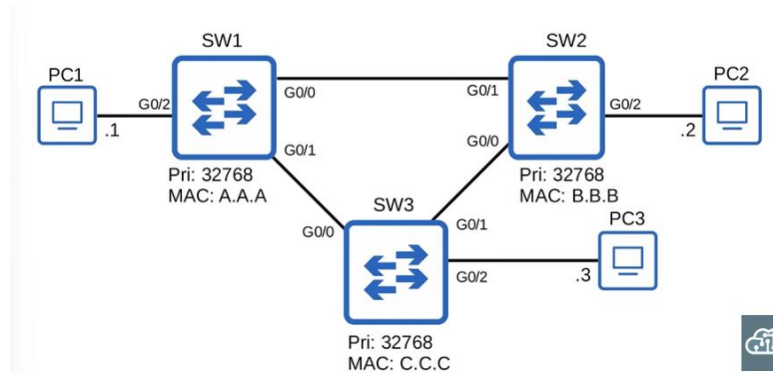


The default bridge priority is 32768.

Since all switches have the same default priority, the MAC address is used as the tie-breaker.

The switch with the lowest MAC address wins.

EXAMPLE TOPOLOGY WITH PRIORITY AND MAC



SW1: Priority 32768, MAC A.A.A

SW2: Priority 32768, MAC B.B.B

SW3: Priority 32768, MAC C.C.C

All switches have the same priority.

MAC addresses are compared.

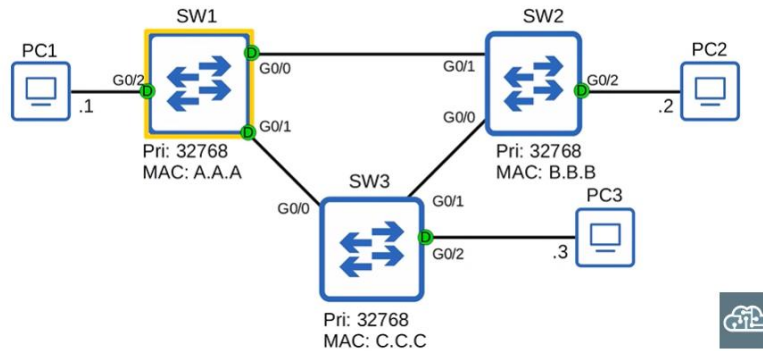
Hexadecimal values:

- A = 10
- B = 11
- C = 12

SW1 has the lowest MAC address.

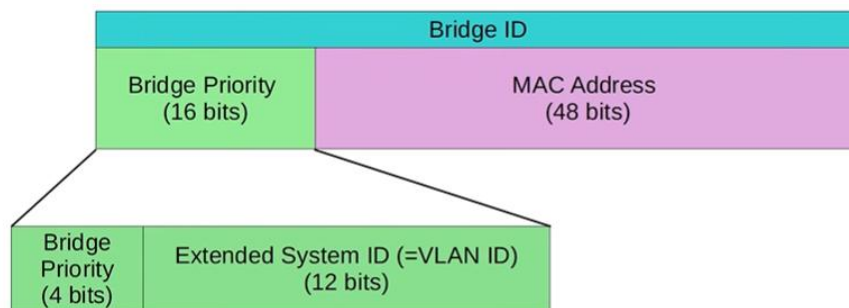
Therefore, SW1 becomes the root bridge.

All ports on SW1 are designated ports and are forwarding.



◆ UPDATED BRIDGE ID (MODERN FORMAT)

The Bridge ID was later updated.



The Extended System ID is the VLAN ID.

WHY VLAN ID IS INCLUDED (PVST)

Cisco uses PVST (Per-VLAN Spanning Tree).

Each VLAN runs its own STP instance.

Because the VLAN ID is included:

- Each VLAN has a different Bridge ID
- A switch can be root for one VLAN but not another

Example:

- VLAN 1 → SW1 is root
- VLAN 2 → SW2 is root
- VLAN 3 → SW3 is root

DEFAULT BRIDGE PRIORITY CALCULATION

Default bridge priority was **32768**, but now the VLAN ID is added.

Default VLAN is VLAN 1.

So the real default priority is:

Bridge Priority				Extended System ID (VLAN ID)														
32768	16384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

$$32768 + 1 = 32769$$

This is why you see **32769** instead of 32768 in practice.

BRIDGE PRIORITY CHANGE RULE (IMPORTANT)

You cannot change the VLAN ID portion.

So the total bridge priority can only change in steps of **4096**.

Valid bridge priority values:

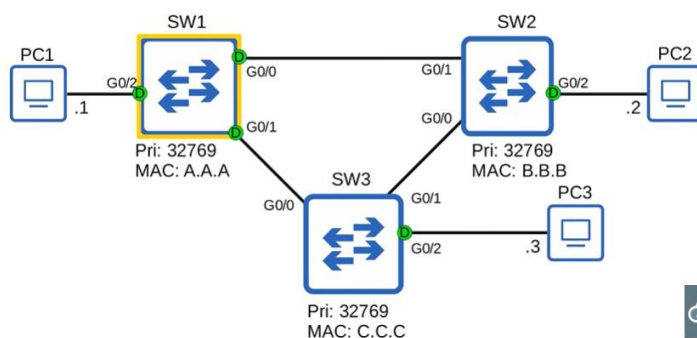
The STP bridge priority can only be changed in units of 4096.

The valid values you can configure are:

0, 4096, 8192, 12288, 16384, 20480, 24576, 28672, 32768, 36864, 40960, 45056, 49152, 53248, 57344, or 61440.

The Extended System ID will then be added to this number to make the total bridge priority.

EXAMPLE PRIORITY REDUCTION



Current total priority (VLAN 1):

$$32768 + 1 = 32769$$

Next lower valid value:

$$16384 + 8192 + 4096 + 1 = 28673$$

This makes the switch more likely to become the root bridge.

◆ ROOT BRIDGE BEHAVIOR

When a switch powers on:

- It assumes it is the root bridge
- It sends BPDUs claiming root status

If it receives a superior BPDUs (lower Bridge ID):

- It gives up being root

After convergence:

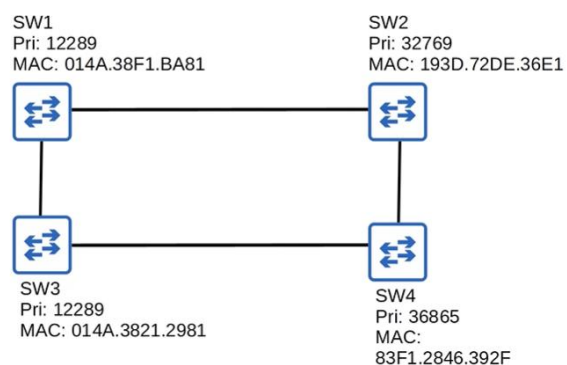
- Only the root bridge originates BPDUs
- Other switches forward BPDUs but do not create new ones

◆ PORT STATES AND LOOP PREVENTION

STP places ports into different states:

- Forwarding
Sends and receives normal traffic
- Blocking
Does not forward user traffic
Only processes BPDUs and specific control traffic

◆ PRACTICE QUESTION 1 – ROOT BRIDGE



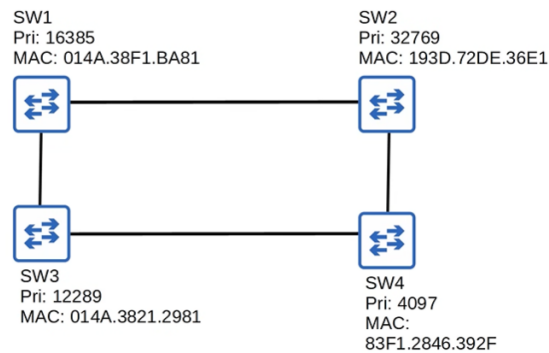
Priorities are equal.

Compare MAC addresses.

SW3 has the lower MAC address.

SW3 becomes the root bridge.

◆ PRACTICE QUESTION 2 – ROOT BRIDGE



Lowest priority wins.

SW4 becomes the root bridge.

◆ SPANNING TREE PROTOCOL

STEP 1: ROOT BRIDGE ELECTION (RECAP)

- The switch with the lowest Bridge ID becomes the root bridge
- Bridge ID = Bridge Priority + Extended System ID (VLAN ID) + MAC address
- If priority is same, lowest MAC address wins
- ALL ports on the root bridge are designated ports
- Designated ports are in forwarding state

STEP 2: ROOT PORT SELECTION

- Every non-root switch selects ONE root port
- The interface with the lowest root cost will be the root port.
- Root port = best path to root bridge
- Root ports are always forwarding

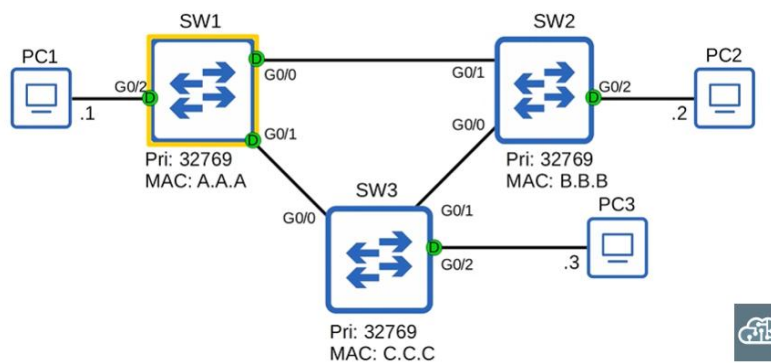
◆ PATH COST VALUES (EXAM IMPORTANT)

- | | Speed | Cost |
|---|------------|------|
| • | 10 Mbps → | 100 |
| • | 100 Mbps → | 19 |
| • | 1 Gbps → | 4 |
| • | 10 Gbps → | 2 |

(All interfaces here are 1 Gbps → cost 4)

◆ ROOT COST CALCULATION

- Root bridge advertises cost = 0
- Each switch adds its outgoing interface cost



COST ADVERTISEMENT FLOW

SW1 (Root)

Cost = 0

SW2:

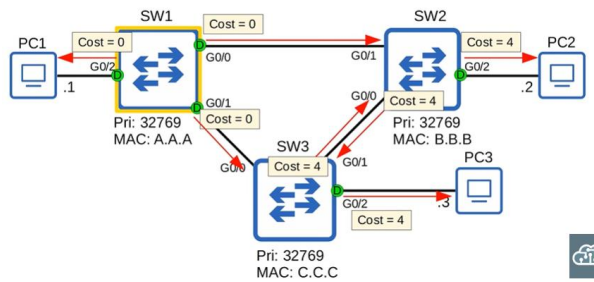
Via G0/1 → $0 + 4 = 4$

Via G0/0 → $4 + 4 = 8$

SW3:

Via G0/0 → $0 + 4 = 4$

Via G0/1 → $4 + 4 = 8$



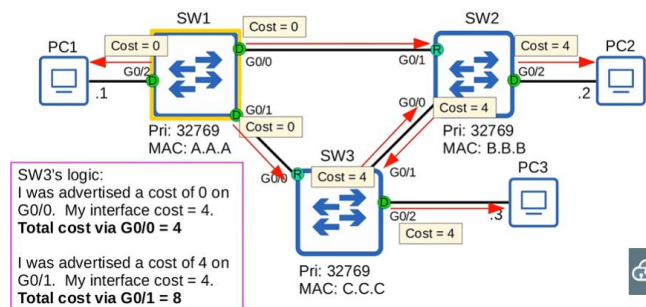
ROOT PORT DECISION

- SW2 root port → **G0/1 (cost 4)**
- SW3 root port → **G0/0 (cost 4)**

ROOT PORTS:

SW2 → G0/1

SW3 → G0/0



IMPORTANT RULE

- The port connected to a root port must be designated
- Root ports must never be blocked

CURRENT PORT STATES

SW1 (ROOT):

G0/0 → Designated (FWD)

G0/1 → Designated (FWD)

SW2:

G0/1 → Root Port (FWD)

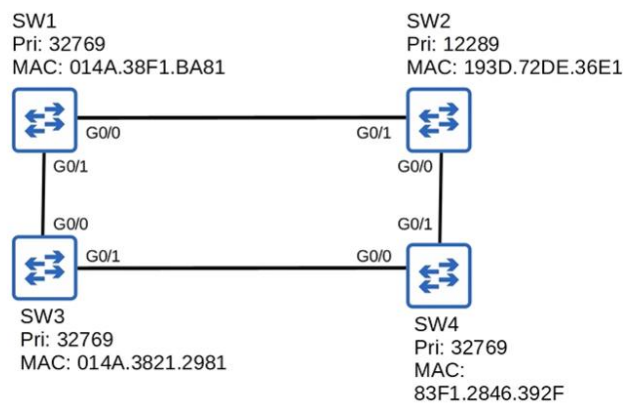
SW3:

G0/0 → Root Port (FWD)

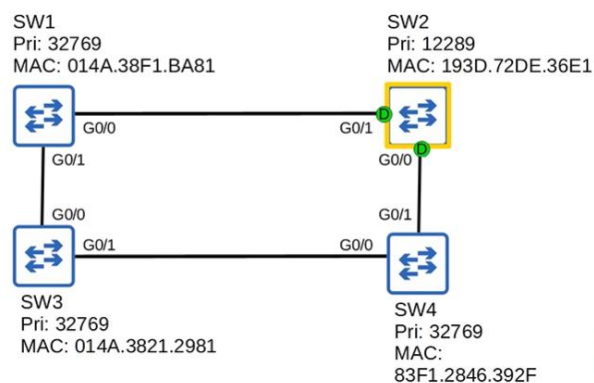
◆ Root Port Selection Rules (Order)

1. Lowest **root cost**
2. If tie → neighbour with **lowest bridge ID**
3. If tie → neighbour's **lowest port ID**

Example: Tie in Root Cost

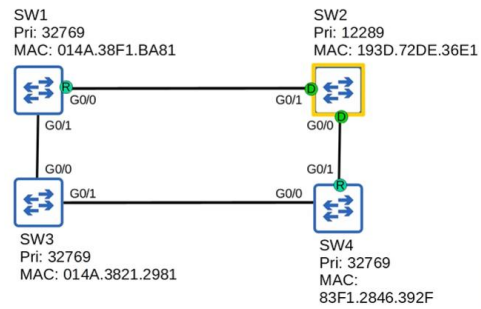


- All interfaces cost = 4
- Root bridge = **SW2** (lowest priority)



Root Ports

- SW1 → G0/0 (cost 4)
- SW4 → G0/1 (cost 4)



SW3 Tie Case

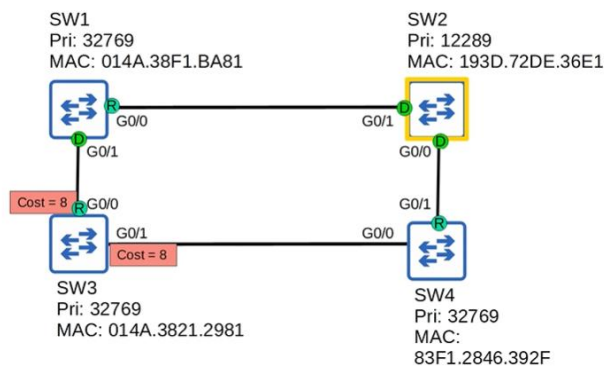
- Via G0/0 → cost = 8
- Via G0/1 → cost = 8

Tie breaker:

- Neighbor bridge ID:
 - SW1 < SW4

Result:

- SW3 selects **G0/0** as root port
- SW1's G0/1 becomes designated



Final Root Port Tie-Breaker: Port ID

- Used when:
 - Root cost is same
 - Neighbour bridge ID is same

Port ID

- STP Port ID = **Port Priority + Port Number**
- Default priority = **128**

```
SW1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     aaaa.aaaa.aaaa
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     aaaa.aaaa.aaaa
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15 sec

Interface    Role Sts Cost      Prio.Nbr  Type
-----
Gi0/0        Desg FwD 4      128.1     Shr
Gi0/1        Desg FwD 4      128.2     Shr
Gi0/2        Desg FwD 4      128.3     Shr
Gi0/3        Desg FwD 4      128.4     Shr
Gi1/0        Desg FwD 4      128.5     Shr
Gi1/1        Desg FwD 4      128.6     Shr
Gi1/2        Desg FwD 4      128.7     Shr
Gi1/3        Desg FwD 4      128.8     Shr
```

- Example:
 - Gi0/0 → 128.1
 - Gi0/1 → 128.2

Prio.Nbr

128.1

128.2

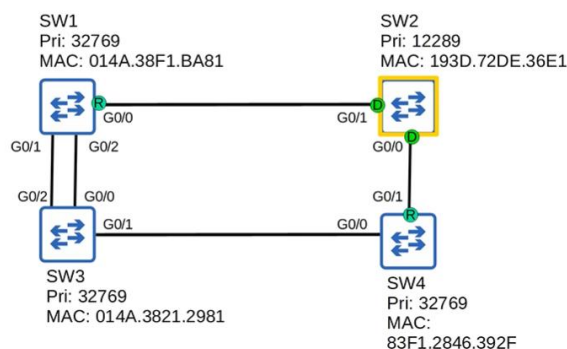
128.3

- Lower port number = lower port ID

Important Point

- Neighbour switch's port ID is used
- NOT the local switch's port ID

Example: Two Links Between SW1 and SW3



- Both links have same root cost

- Same neighbour bridge ID
- Tie-breaker now we compare neighbour lower port id

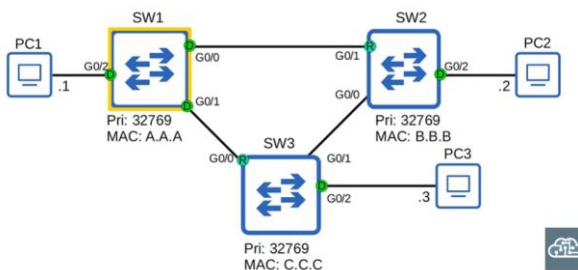
Result:

- SW3 selects the port connected to **lower port ID on SW1**
- Root port = **G0/2**
- SW1's connected port becomes designated

◆ Designated and Blocking Ports

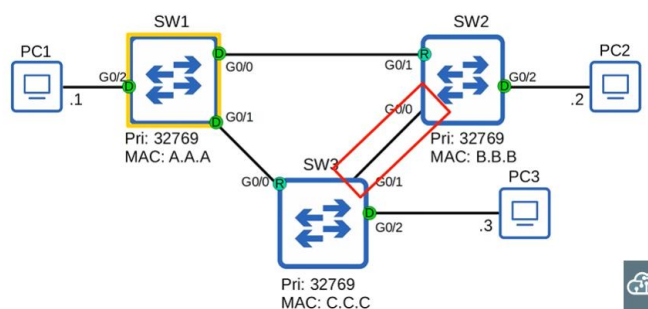
- Root ports → Forwarding
- Designated ports → Forwarding
- To prevent loops → some ports must block

◆ NOW FINAL TOPOLOGY



Current Situation:

- All that's left is the connection between SW2 and SW3



- So far, all ports are in a forwarding state
- Root ports and designated ports are always in a forwarding state

◆ Important Rule

Every collision domain has a single spanning tree designated port

- When we use switches, each link is a separate collision domain
- PCs don't participate in spanning tree

◆ Collision Domains and Designated Ports

SW1 ↔ SW2

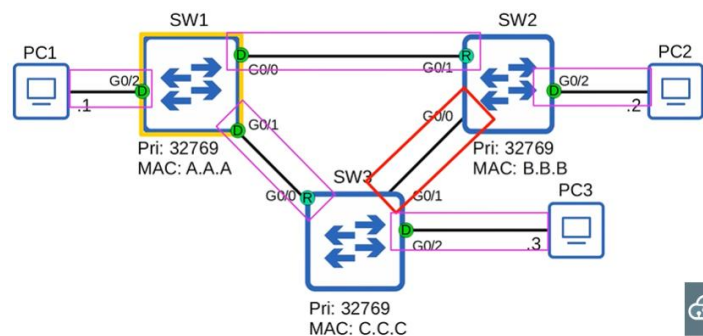
- This collision domain has one designated port
- SW1's G0/0

SW1 ↔ SW3

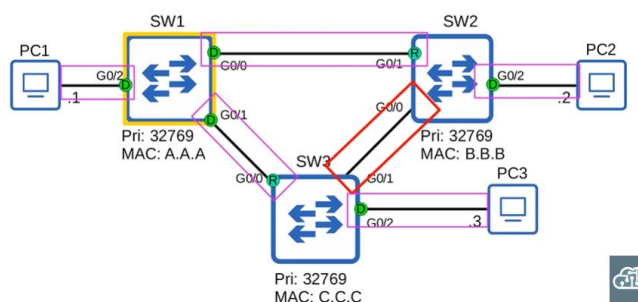
- This connection has one designated port
- SW1's G0/1

Switch ↔ PC Links

- All connections with the PCs are designated ports
- They are in the forwarding state



◆ Remaining Collision Domain: SW2 ↔ SW3



- We need one designated port on this connection
- Do we block both ports?
✗ No

How Is the Designated Port Chosen?

Rule 1: The switch with the lowest root cost will make its port designated

- SW2 root cost = **4** (via G0/1)
- SW3 root cost = **4** (via G0/0)
- Root cost is the **same**

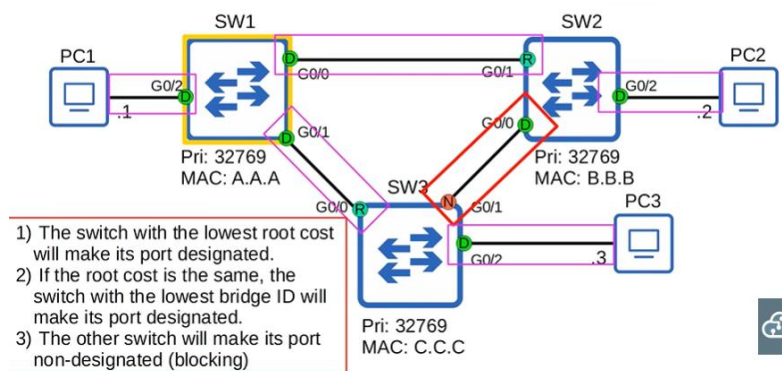
Rule 2 (Tie-breaker): Compare the bridge ID

- SW2 has the lower bridge ID
- So SW2's G0/0 is designated

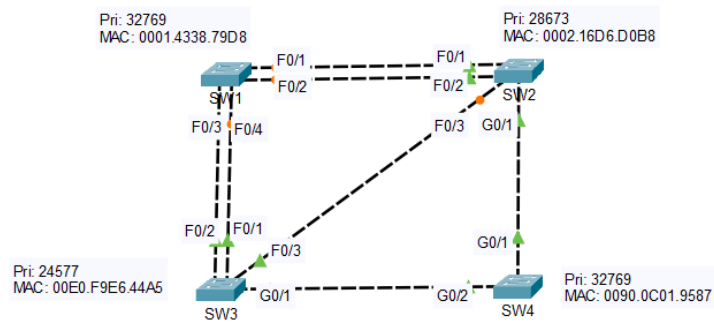
Rule 3

The other switch will make its port non-designated

- SW3's G0/1 is non-designated
- It is in a blocking state
- This blocks the port to prevent Layer 2 loop



◆ Some Command of Spanning Tree Protocol



AT SW3 Information:

1] “show spanning-tree”

```
SW3#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
            Address     00E0.F9E6.44A5
            This bridge is the root
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
            Address     00E0.F9E6.44A5
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/3        Desg FWD 19        128.3    P2p
Fa0/2        Desg FWD 19        128.2    P2p
Fa0/1        Desg FWD 19        128.1    P2p
Gi0/1        Desg FWD 4        128.25   P2p
```

2] “show spanning-tree vlan 1”

```
SW3#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
            Address     00E0.F9E6.44A5
            This bridge is the root
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
            Address     00E0.F9E6.44A5
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/3        Desg FWD 19        128.3    P2p
Fa0/2        Desg FWD 19        128.2    P2p
Fa0/1        Desg FWD 19        128.1    P2p
Gi0/1        Desg FWD 4        128.25   P2p

SW3#
```


3] “show spanning-tree detail”

```
SW3#show spanning-tree detail

VLAN0001 is executing the ieee compatible Spanning Tree Protocol
Bridge Identifier has priority of 24576, sysid 1, 00E0.F9E6.44A5
Configured hello time 2, max age 20, forward delay 15
We are the root of the spanning tree
Topology change flag not set, detected flag not set
Number of topology changes 0 last change occurred 00:00:00 ago
    from FastEthernet0/1
Times: hold 1, topology change 35, notification 2
    hello 2, max age 20, forward delay 15
Timers: hello 0, topology change 0, notification 0, aging 300

Port 1 (FastEthernet0/1) of VLAN0001 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.1
Designated root has priority 24577, address 00E0.F9E6.44A5
Designated bridge has priority 24577, address 00E0.F9E6.44A5
Designated port id is 128.1, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is point-to-point by default

Port 2 (FastEthernet0/2) of VLAN0001 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.2
Designated root has priority 24577, address 00E0.F9E6.44A5
Designated bridge has priority 24577, address 00E0.F9E6.44A5
Designated port id is 128.2, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is point-to-point by default
```

4] “show spanning-tree summary”

```
SW3#show spanning-tree summary
Switch is in pvst mode
Root bridge for: default
Extended system ID          is enabled
Portfast Default            is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default           is disabled
EtherChannel misconfig guard is disabled
UplinkFast                  is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	4	4
1 vlans	0	0	0	4	4

◆ **Complete Spanning Tree Process (From the Video)**

1. One switch is selected as the root bridge
 - The switch with the lowest bridge ID
2. Each remaining switch selects ONE root port
 - Lowest root cost
 - If tie → neighbour with lowest bridge ID
 - If tie → lowest port ID on the neighbour switch
3. Each remaining collision domain selects ONE designated port
 - Lowest root cost
 - If tie → lowest bridge ID
 - The other port is non-designated (blocking)